

Eray Mutlu

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PROFILE

Biomedical engineer specializing in physiological signal analysis for digital health. Research experience includes ear-EEG-based driver drowsiness detection, explainable AI for ECG, and embedded auscultation systems. First-author journal and conference publications. Experienced in biosignal preprocessing, deep learning, human-subject studies, and industry-academia collaboration.

RESEARCH INTERESTS

Neural signal acquisition and processing; cognitive and physiological state inference; deep learning for biosignals; explainable AI for neural and cardiac signals; neurotechnology.

EDUCATION

Technical University of Denmark (DTU)

M.Sc. in Biomedical Engineering

Lyngby, Denmark

2024 – 2026

- Graduation thesis: Early Prediction of Driver Drowsiness Using Ear-EEG. (Grade: 12/12)

Yıldız Technical University

B.Sc. in Biomedical Engineering

Istanbul, Turkey

2018 – 2023

- Graduation thesis: Digital Stethoscope Design Using TinyML. (Grade: 4/4)

PROFESSIONAL EXPERIENCE

ITU Electro-Optic Devices Laboratory

Undergraduate Researcher

Istanbul, Turkey

Jul – Sep 2023

- Investigated optical triangulation for imaging sulcus vocalis; designed components in SolidWorks and 3D-printed prototypes; simulated optical paths in ZEMAX.

Medical Imaging Academy (MIA)

Biomedical Engineer (Intern)

Istanbul, Turkey

Jul 2022

- Tested RF coils, filters and amplifiers for MRI; assisted in CT system installation/calibration; gained hands-on exposure to MRI/CT/angiography/ultrasound systems.

RESEARCH PROJECTS

Driver Drowsiness Detection via Ear-EEG

2024 – Present

- Industry-academia collaboration with Cebro Medical; ethics-approved human ear-EEG study.

Explainable AI for Electrocardiography

2025 – Present

- Counterfactual methods for interpretable ECG classification (single-lead, single-beat).

Digital Stethoscope Using TinyML

2022 – 2023

- Embedded auscultation system.

PUBLICATIONS

Mutlu, E.*, Shen, D.*, Hempel, P., Brink-Kjær, A., You, L., & Spicher, N. *Interpreting AI ECG Models via Physiology-Inspired Counterfactual Explanations*. Accepted at IEEE EMBC, 2026.

Mutlu, E., Saibene, M., de Araujo, J., Ahrens, E., & Puthusserypady, S. *Clustering-Based Detection of Driver Drowsiness with Ear-EEG*. 14th International Conference on Brain-Computer Interface (BCI), 2026.

Mutlu, E., Hüseyin, V., & Serbes, G. *Bridging Auscultation and Tiny Machine Learning*. Traitement du Signal, 2024.

Mutlu, E., Hüseyin, V., & Serbes, G. *FISC-Net: Classification of Auscultation Sounds on an Edge Device*. 2023 IEEE IMSA.

SKILLS & RESEARCH PRACTICE

Biosignal Analysis: Data-driven analysis and modeling of neural and cardiac signals, integrating signal processing, machine learning, and physiological interpretation.

Experimental & Systems Experience: Human-subject data acquisition; design and deployment of embedded physiological sensing system.

Research Practice: Ethics and funding applications, experimental protocol design, scientific writing, and industry-academia collaboration.

AWARDS & FUNDING

TUBITAK 2209-A Undergraduate Research Grant <i>The Scientific and Technological Research Council of Türkiye</i>	2023
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VOLUNTEERING ACTIVITIES

In Vivo Neuroscience Workshop, 2026 — In vivo neural recording and manipulation, including optical imaging, fiber photometry, electrophysiology, Neuropixels, neural data analysis, and behavioral paradigms.

BCI & Neurotechnology Spring School (g.tec), 2023 — Motor imagery EEG for stroke rehabilitation hackathon.

IEEE YTU Student Branch, 2020–2021 — Obstacle-avoiding robot with Arduino.

LANGUAGES

Turkish (Native), English (Fluent), Danish (Beginner).